HOSPITALITY • REVENUE ANALYTICS

PRE-HACKATHON BRIEF • 13 JUNE 2026

Hotel Revenue Prediction

A Linear Regression Case Study

Predicting Average Daily Rate from Real Portuguese Hotel Booking Data

119,390	32	2015–2017	177
Bookings	Features	Date Range	Countries

Dataset Source:

Antonio, N., de Almeida, A., & Nunes, L. (2019). Hotel booking demand datasets.

Data in Brief, 22, 41-49. Elsevier. |

About This Document

This document is your pre-hackathon briefing pack. It contains everything you need to understand the problem, the domain context, and the dataset — so you arrive on 13 June 2026 ready to build, not read.

The actual dataset will be distributed on Hackathon Day. Use this document to familiarise yourself with the columns, data quality issues, and business context.

1. Domain Background

The global hotel industry is a multi-trillion dollar market where pricing decisions are made thousands of times per day. Unlike traditional retail, hotel rooms are a perishable product — an unoccupied room on any given night represents revenue that can never be recovered. This fundamental constraint has driven the hospitality sector to become one of the most sophisticated adopters of data-driven revenue management.

At the centre of hotel revenue management sits the **Average Daily Rate (ADR)** — the single most-watched pricing metric in the industry. ADR is not set arbitrarily; it is the outcome of a complex interplay between demand signals, booking channel economics, customer segment behaviour, seasonality, and competitive positioning.

The Revenue Management Ecosystem

Concept	What it means	Relevance to this dataset
ADR	Total room revenue ÷ total room nights sold. The primary pricing KPI.	This is your target variable (Y)
RevPAR	Revenue Per Available Room = ADR × Occupancy Rate.	ADR is the controllable lever in the RevPAR equation
Lead Time	Days between booking and arrival. Longer lead time = different demand segment.	lead_time column — key predictor
Distribution Channel	How the booking reached the hotel: OTA, direct, corporate, GDS.	market_segment & distribution_channel columns
Seasonal Demand	Hotels adjust rates for peak/off-peak periods (summer, holidays, events).	arrival_date_month, week_number columns
Deposit Policy	Non-refundable rates are typically cheaper; flexible rates command a premium.	deposit_type column — pricing signal

About the Hotels

The dataset covers two real, anonymised hotels located in Portugal — a country that receives over 20 million international tourists annually and whose hospitality sector is dominated by leisure and city break travel.

	City Hotel	Resort Hotel
Location	Lisbon metropolitan area	Algarve coast (southern Portugal)
Guest profile	Business & short-stay leisure	Leisure & holiday families
Bookings in dataset	79,330 (66.5%)	40,060 (33.5%)
Typical stay	1–3 nights	3–7 nights
Peak season	Year-round (business), summer peaks	June–September (beach season)

2. Problem Statement

Objective	Build a **multiple linear regression model** that predicts the **Average Daily Rate (ADR, in Euros)** of a hotel booking, using only information that would be available *at the time of booking*.

The Target Variable: ADR

ADR is calculated as:

ADR = Total Room Revenue for a Booking ÷ Total Room Nights

A booking for 3 nights at a total room cost of €300 has an ADR of €100. The model must learn what booking-level factors drive this per-night rate — from the type of hotel and booking channel, to the meal plan and arrival season.

Scope & Constraints

In scope	Predicting ADR from pre-booking attributes only
In scope	Feature engineering from the 32 available columns
In scope	Applying linear regression with appropriate preprocessing
Out of scope	Predicting booking cancellations (classification, not regression)
Out of scope	Using post-booking information (reservation_status, status_date)
Out of scope	Forecasting hotel-level revenue totals (aggregation task)

Business Questions to Explore

As you build your model, use it to answer the following questions:

Q1	Which booking attributes have the strongest influence on the nightly room rate?
Q2	Does the booking channel (Direct vs Online TA vs Corporate) significantly affect ADR?
Q3	How does seasonality — arrival month and week number — impact pricing?
Q4	Do resort hotels command a higher ADR than city hotels after controlling for other factors?
Q5	What is the expected ADR range for a new booking with a given set of characteristics?

3. Dataset Description

The dataset contains **119,390 booking records** across **32 columns**, sourced from two real Portuguese hotels. Each row is a single booking. The table below describes every column — its data type, meaning, and modelling notes.

Target variable (Y)	Data leakage — must not be used as a feature	Requires engineering / special handling

Column	Type	Description & Modelling Notes
`hotel`	str	Type of hotel: **City Hotel** or **Resort Hotel**. Key categorical split — resort properties have distinct seasonal pricing patterns.
`is_canceled`	int	1 = booking cancelled, 0 = completed. Filter to non-cancelled bookings before modelling revenue; cancelled ADR values may be indicative, not final.
`lead_time`	int	Days between booking creation and arrival. Range: 0–737. Early-bird bookings may be made at different rates than last-minute ones.
`arrival_date_year`	int	Year of arrival: 2015, 2016, or 2017. Useful for capturing year-on-year pricing inflation.
`arrival_date_month`	str	Month of arrival as a string (January – December). Strong seasonal pricing signal — encode numerically (1–12) or with cyclical sin/cos encoding.
`arrival_date_week_number`	int	ISO week number (1–53). Finer seasonal granularity than month; captures peak summer weeks in the Portuguese tourist calendar.
`arrival_date_day_of_month`	int	Day of month (1–31). Limited direct pricing effect but useful for weekend/weekday interaction features.
`stays_in_weekend_nights`	int	Weekend nights (Saturday/Sunday) in the stay. Combine with weekday nights to engineer **total_nights**.
`stays_in_week_nights`	int	Weekday nights (Monday–Friday) in the stay. A booking with both = 0 indicates a day-use reservation.
`adults`	int	Number of adult guests. Combine with children and babies to engineer **total_guests** (occupancy proxy).
`children`	float	**4 missing values** — impute with 0 (most bookings have no children). Can be used as a numeric count or binary flag.
`babies`	int	Number of infants. Mostly 0; contributes to total occupancy. Rarely a direct pricing driver.
`meal`	str	Meal plan: **BB** (Bed & Breakfast), **HB** (Half Board), **FB** (Full Board), **SC** (Self-Catering), **Undefined**. Encode as dummy variables.
`country`	str	Guest country of origin (ISO 3166-1 alpha-3). **488 missing values. 177 unique levels** — too many for naive OHE. Group into top-N + "Other" or use target encoding.
`market_segment`	str	Booking channel category: Online TA, Offline TA/TO, Direct, Corporate, Groups, Complementary, Aviation, Undefined. One of the most impactful categorical predictors.
`distribution_channel`	str	How the booking reached the hotel: TA/TO, Direct, Corporate, GDS, Undefined. Partially overlaps with market_segment — check for multicollinearity before including both.
`is_repeated_guest`	int	1 = returning guest, 0 = new. Loyal guests may receive discounted rates — negatively correlated with ADR (r = –0.13).

Column	Type	Description & Modelling Notes
`previous_cancellations`	int	Number of bookings previously cancelled by this customer. Acts as a risk/trust signal.
`previous_bookings_not_canceled`	int	Count of prior completed bookings by this customer. Proxy for customer loyalty and engagement level.
`reserved_room_type`	str	Room category originally requested (anonymised A–L). Engineer a **room_mismatch** flag when this differs from assigned_room_type.
`assigned_room_type`	str	Room category actually assigned at check-in. May differ due to hotel overbooking, upgrades, or operational constraints.
`booking_changes`	int	Number of amendments made to the booking before arrival. Indicates engaged customers or complex itineraries.
`deposit_type`	str	Deposit policy at booking: **No Deposit** (87.6%), **Non Refund** (12.2%), **Refundable** (0.1%). Encodes price-commitment and demand signal.
`agent`	float	Travel agent ID. **16,340 missing (13.7%)** — treat missing as "no agent" by creating a binary **has_agent** flag. Drop the raw ID column.
`company`	float	Corporate account ID. **112,593 missing (94.3%)** — create a binary **is_corporate** flag. The raw column has no predictive value as-is.
`days_in_waiting_list`	int	Days the booking spent on a waiting list before confirmation. Mostly 0; high values indicate high-demand periods or constrained supply.
`customer_type`	str	Customer classification: **Transient** (75%), **Transient-Party** (21%), **Contract** (3%), **Group** (<1%). Strong pricing segment indicator.
`adr`	float	**TARGET VARIABLE (Y).** Average Daily Rate in Euros. Mean: €101.83 | Median: €94.58 | Std: €50.54. Contains 1,959 zero values (complimentary stays) and one extreme outlier (€5,400) — both require treatment.
`required_car_parking_spaces`	int	Parking spaces requested. Proxy for car-travelling guests; higher in resort bookings from domestic markets.
`total_of_special_requests`	int	Number of special requests made (e.g., high floor, cot, late checkout). Range: 0–5. Positive correlation with ADR (r = 0.17) — higher-paying guests make more requests.
`reservation_status`	str	**DATA LEAKAGE — DO NOT USE.** Check-Out / Canceled / No-Show is set after the stay completes, directly encoding post-booking outcome. Using it would inflate model performance artificially.
`reservation_status_date`	str	**DATA LEAKAGE — DO NOT USE.** Date the reservation status was last updated. Post-booking information — must be excluded alongside reservation_status.

4. Data Snapshot

The table below shows a representative sample of 8 rows from the dataset across selected columns. The full dataset (119,390 rows × 32 columns) will be distributed on Hackathon Day.

hotel	year	month	wknd nts	wk nts	adults	meal	market segment	cancelled	ADR (€)
Resort Hotel	2015	July	0	0	2	BB	Direct	0	**0.00**
Resort Hotel	2015	July	0	0	2	BB	Direct	0	**0.00**
Resort Hotel	2015	July	0	1	1	BB	Direct	0	**75.00**
Resort Hotel	2015	July	0	1	1	BB	Corporate	0	**75.00**
Resort Hotel	2015	July	0	2	2	BB	Online TA	0	**98.00**
Resort Hotel	2015	July	0	2	2	BB	Online TA	0	**98.00**
Resort Hotel	2015	July	0	2	2	BB	Direct	0	**107.00**
Resort Hotel	2015	July	0	2	2	FB	Direct	0	**103.00**

Note: 14 of 32 columns shown. Two rows display ADR = 0.00 — these are complimentary bookings and represent a data quality decision point during preprocessing.

5. Data Quality Notes

The table below summarises the key data quality issues you will encounter. Understanding these before Hackathon Day will give you a significant head start.

Issue	Column(s)	Scale	Recommended Approach
Missing values	children	4 rows	Impute with 0 (no children assumed)
Missing values	country	488 rows (0.4%)	Impute as "Unknown" category
Missing values	agent	16,340 rows (13.7%)	Create binary has_agent flag; drop raw column
Near-complete missing	company	112,593 rows (94.3%)	Create binary is_corporate flag; drop raw column
Zero-value target	adr	1,959 rows (1.6%)	Filter out — complimentary stays distort regression
Extreme outlier	adr	1 row (€5,400)	Remove — single anomalous value; max non-cancelled = €510
High cardinality	country	177 unique values	Group top countries + "Other"; or use target encoding
Data leakage	reservation_status reservation_status_date	2 columns	Drop entirely — post-booking outcome columns
Multicollinearity risk	market_segment + distribution_channel	2 columns	Check VIF; consider using only one or create interaction term